Impact of dietary shift to higher-antioxidant foods in COPD: a randomised trial.

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Chronic obstructive pulmonary disease (COPD) is characterised by increased oxidative stress. Dietary factors, such as ample consumption of foods rich in antioxidants, such as fruit and vegetables, might have beneficial effects in COPD patients. The association between dietary shift to foods rich in antioxidants and lung function in COPD was investigated in a 3-yr prospective study. A total of 120 COPD patients were randomised to follow either a diet based on increased consumption of fresh fruit and vegetables (intervention group (IG)) or a free diet (control group (CG)). The mean consumption of foods containing antioxidants was higher in the IG than in the CG throughout the study period ($p<0.05$). The relationship between consumption of foods rich in antioxidants and percentage predicted forced expiratory volume in 1 s was assessed using a general linear model for repeated measures; the two groups overall were different in time ($p = 0.03$), with the IG showing a better outcome. In investigating the effect of several confounders (sex, age, smoking status, comorbid conditions and exacerbation) of group response over time, nonsignificant interactions were found between confounders, group and time. These findings suggest that a dietary shift to higher-antioxidant food intake may be associated with improvement in lung function, and, in this respect, dietary interventions might be considered in COPD management.